CSP-203: Software Tools & Technologies Laboratory



Lab Session - 2

Hands on Basic Linux Commands

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Objectives

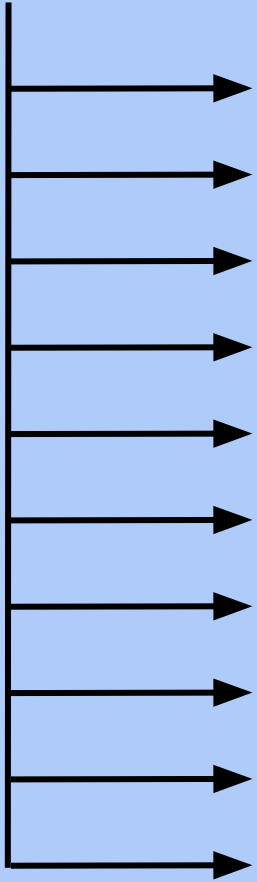
#1
GENERAL USER COMMANDS

#2
FILE SYSTEM AND MANAGEMENT

#3
USER MANAGEMENT AND ADMIN PRIVILEGES

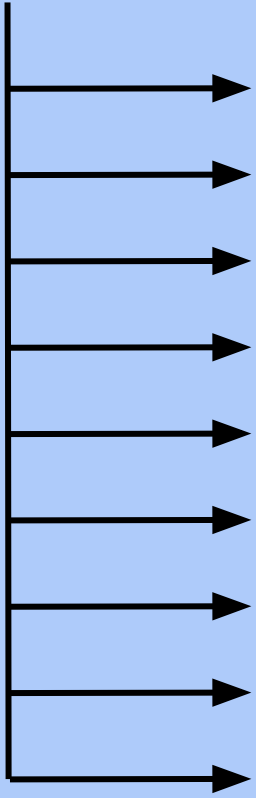
#4
PROCESS MANAGEMENT & TEXT PROCESSING TOOLS

General User Commands



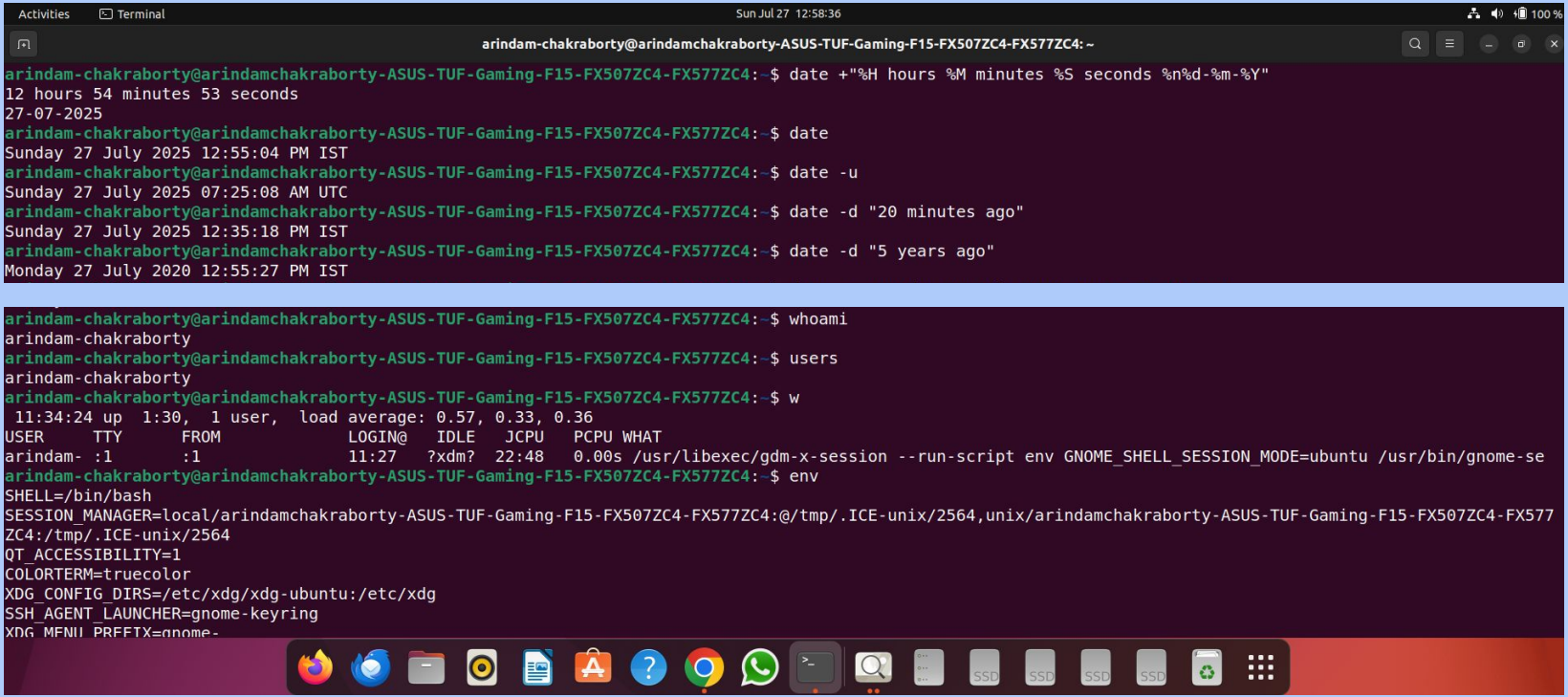
- Opening a terminal in linux: Using GUI / Ctrl + Alt + T Exiting: exit
- See the manual of any command: man command, command --help
- Showing current username: whoami
- Changing current password: passwd
- Who is logged in: w / who
- Show the users logged in to the current host: users
- Print effective user id: whoami
- History: Shows command history / Clear the current terminal screen: clear
- Show the path to the current directory: pwd
- Display the current regional date and time: date

General User Commands

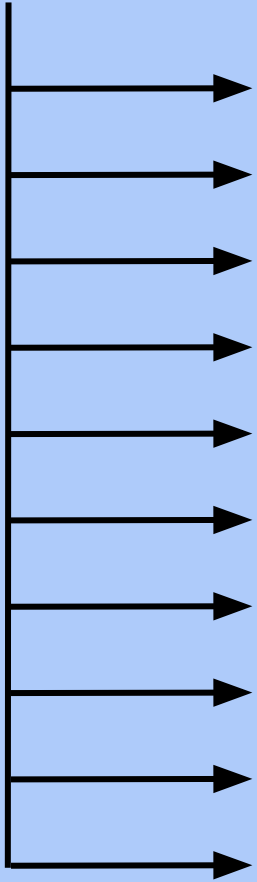


- Show the current universal time and date: date -u
- Display time described by STRING, not 'now': date -d, --date=STRING
- Display current month’s calendar: cal
- Display current, previous and next month: cal -3
- Display calendar of a year: cal year_num
- Display calendar of a given month of a year: cal month_num year_num
- Show environment variables: env
- Print all system information: uname -a
- Printing the date-time in specific format: date +”String_name” [See manual]

General User Commands

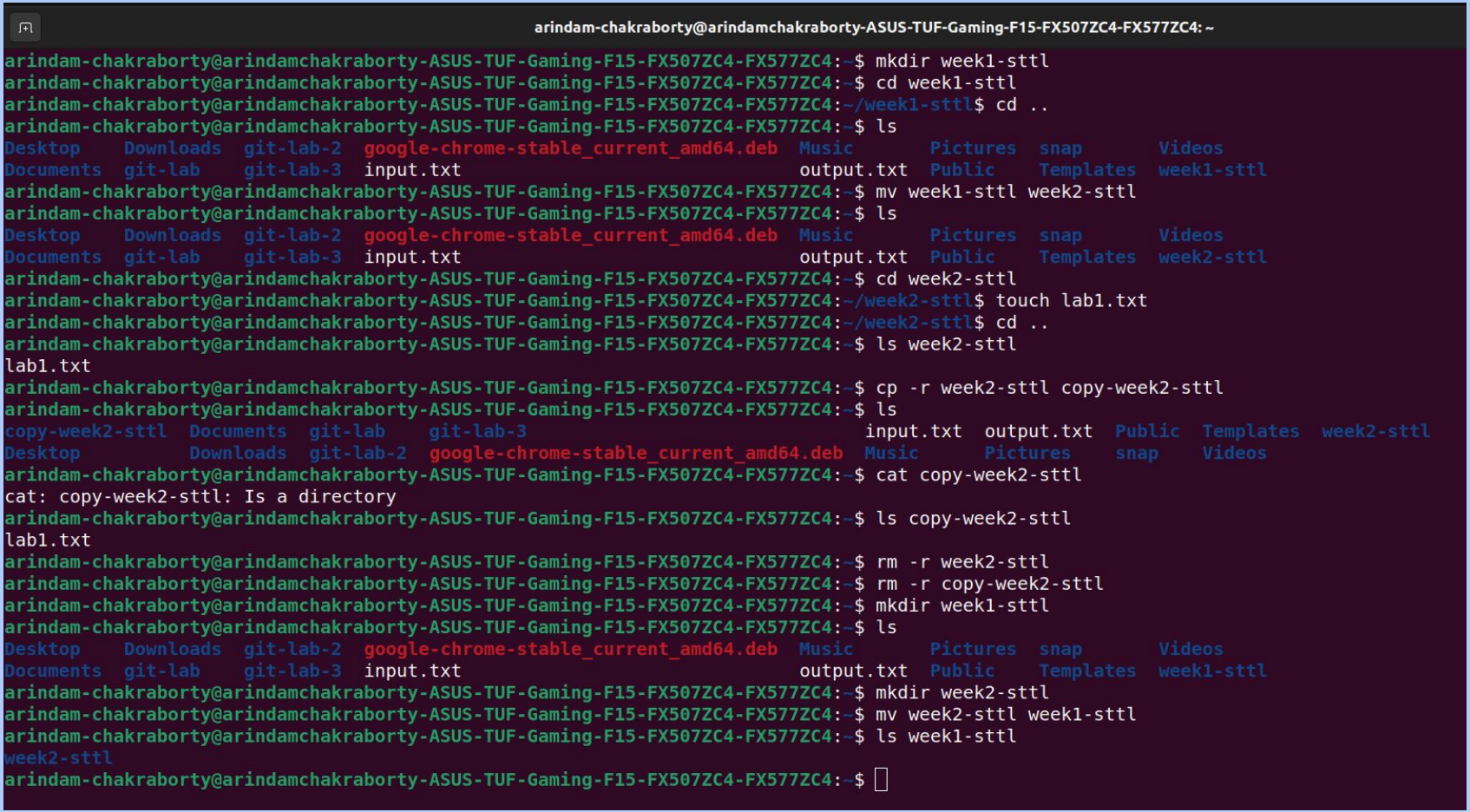


Directory Managements



- Change Directory: cd directory_name [Note: Make sure the directory exists]
- Enter to subdirectory of a directory: cd directory_name/sub_directory_name
- Creating a directory: mkdir new_directory_name
- Returning to parent directory: cd ..
- Returning to home directory: cd
- Deleting an empty directory: rmdir directory_name
- Deleting a non-empty directory: rm -r directory_name
- Renaming a directory: mv old_directory_name new_directory_name
- Moving 1st dir inside 2nd dir: mv first_dir_name second_dir_name
- Copying a directory: cp -r target_directory new_directory

Directory Managements

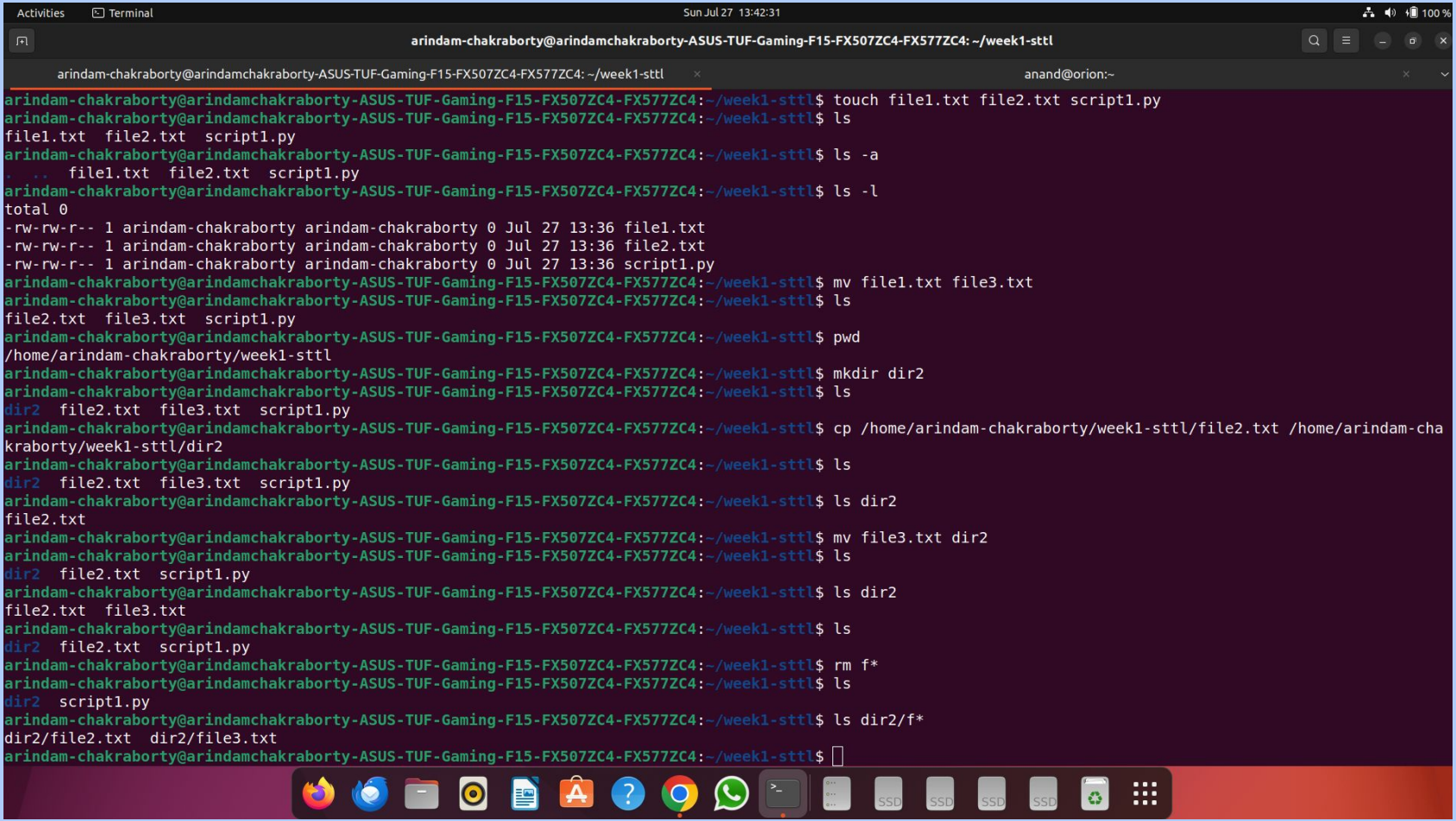


File Managements

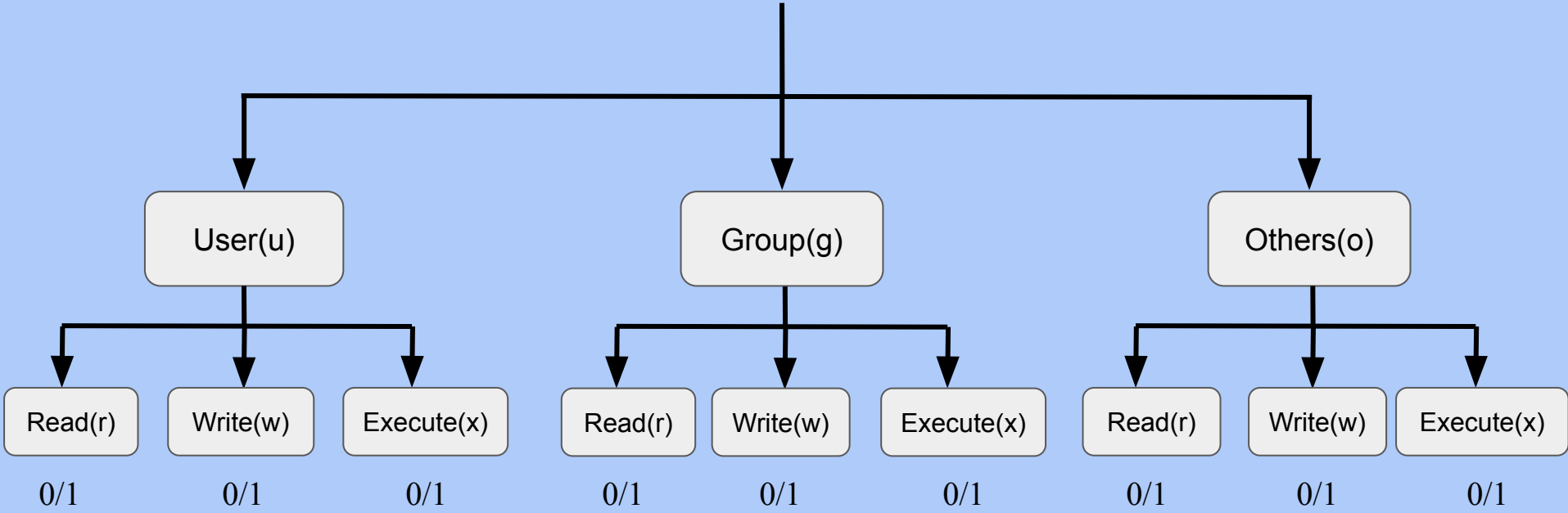


- Creating a file: touch file1 file2 ….. file n
- Deleting a file: rm file_name
- Renaming a file: mv old_file_name new_file_name
- Copying a file to a new directory: cp path_to_file/filename path_to_dest
- Moving a file to new destination: mv path_to_file/filename path_to_dest
- Displaying the list of files and directories within current directory: ls
  - Listing contents of specified directories: ls directory_name
  - Listing contents with details: ls -l
  - Listing hidden contents: ls -a
  - Listing contents ending with .txt: ls *.txt / Starting with a : ls a*

File Managements



File Ownership Management

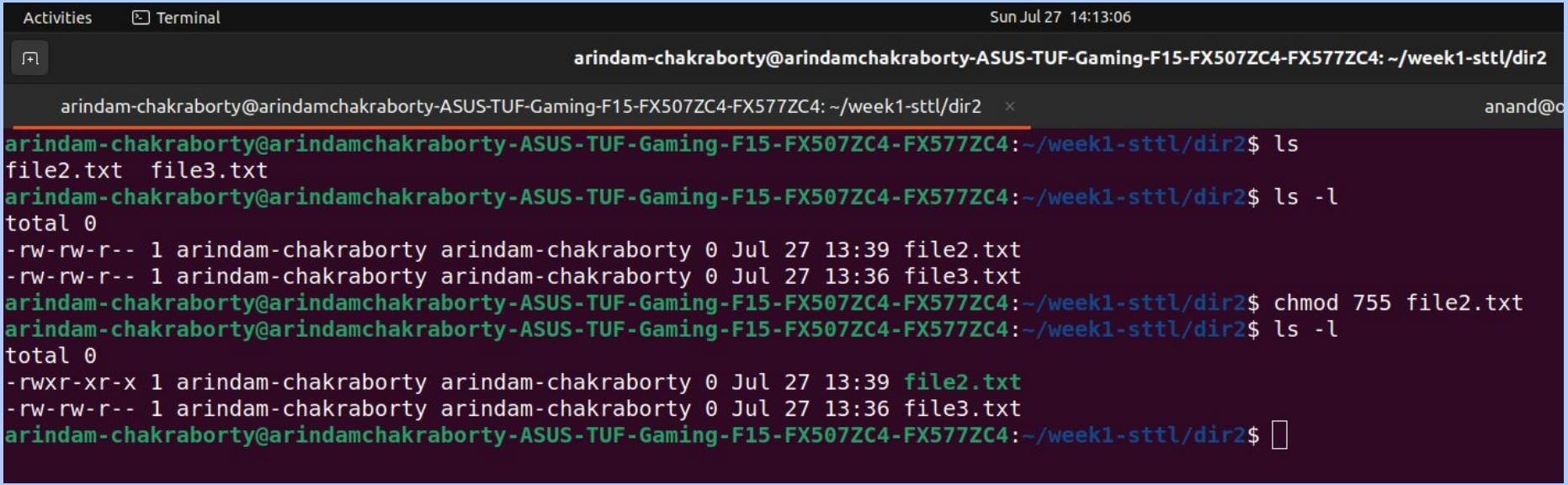


Example: Only read and execution is granted for group. Then the value corresponding to group permission will be (101) : $1*2^2+0*2^1+1*2^0 = 5$

1 -> Granted
0 -> Denied

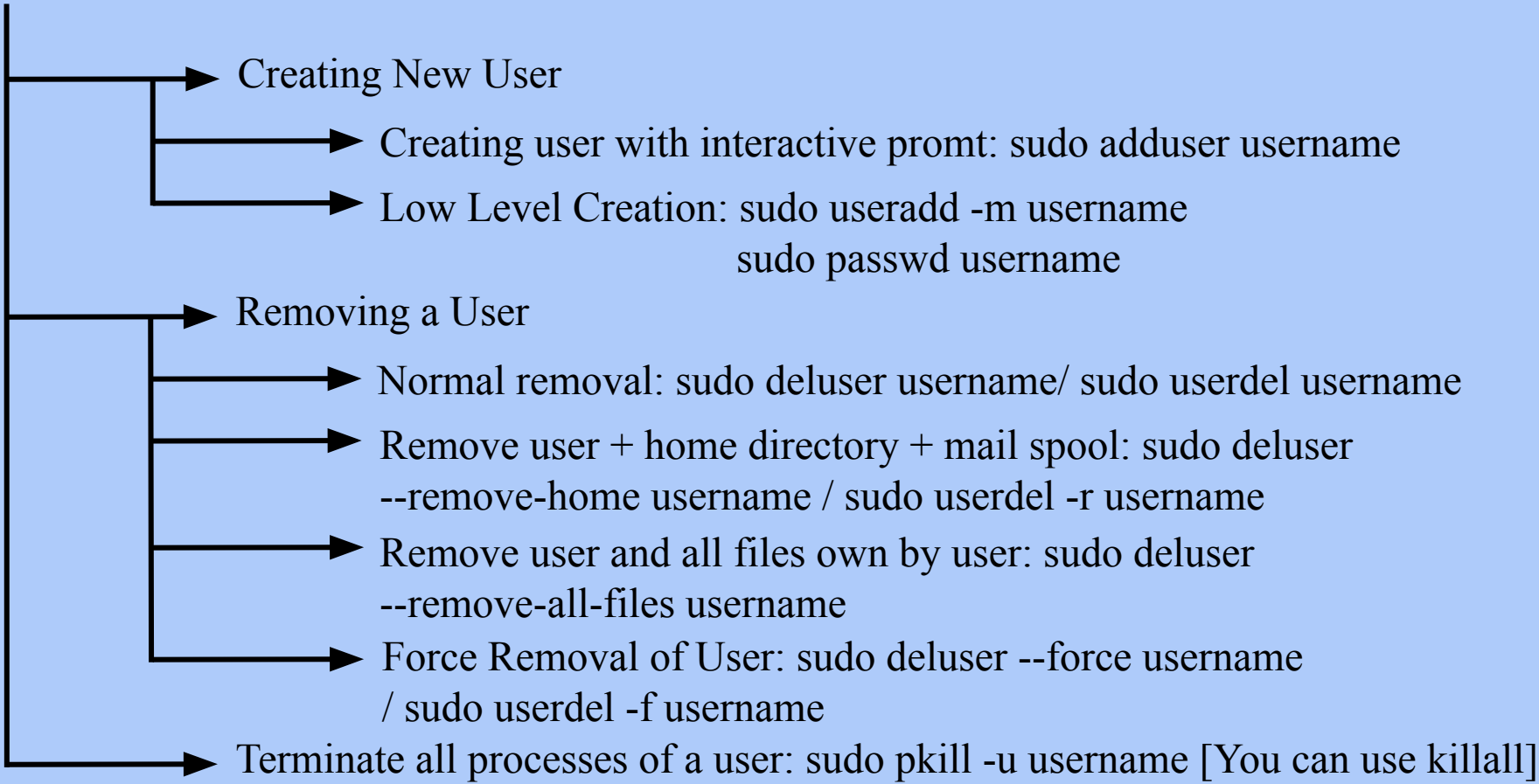
File Ownership Managements

chmod <mode> <file>
Mode: [0-7][0-7][0-7] (for user, group, others)
Example: chmod 755 file2.txt / chmod u=rwx, g=rx, o=rx file2.txt

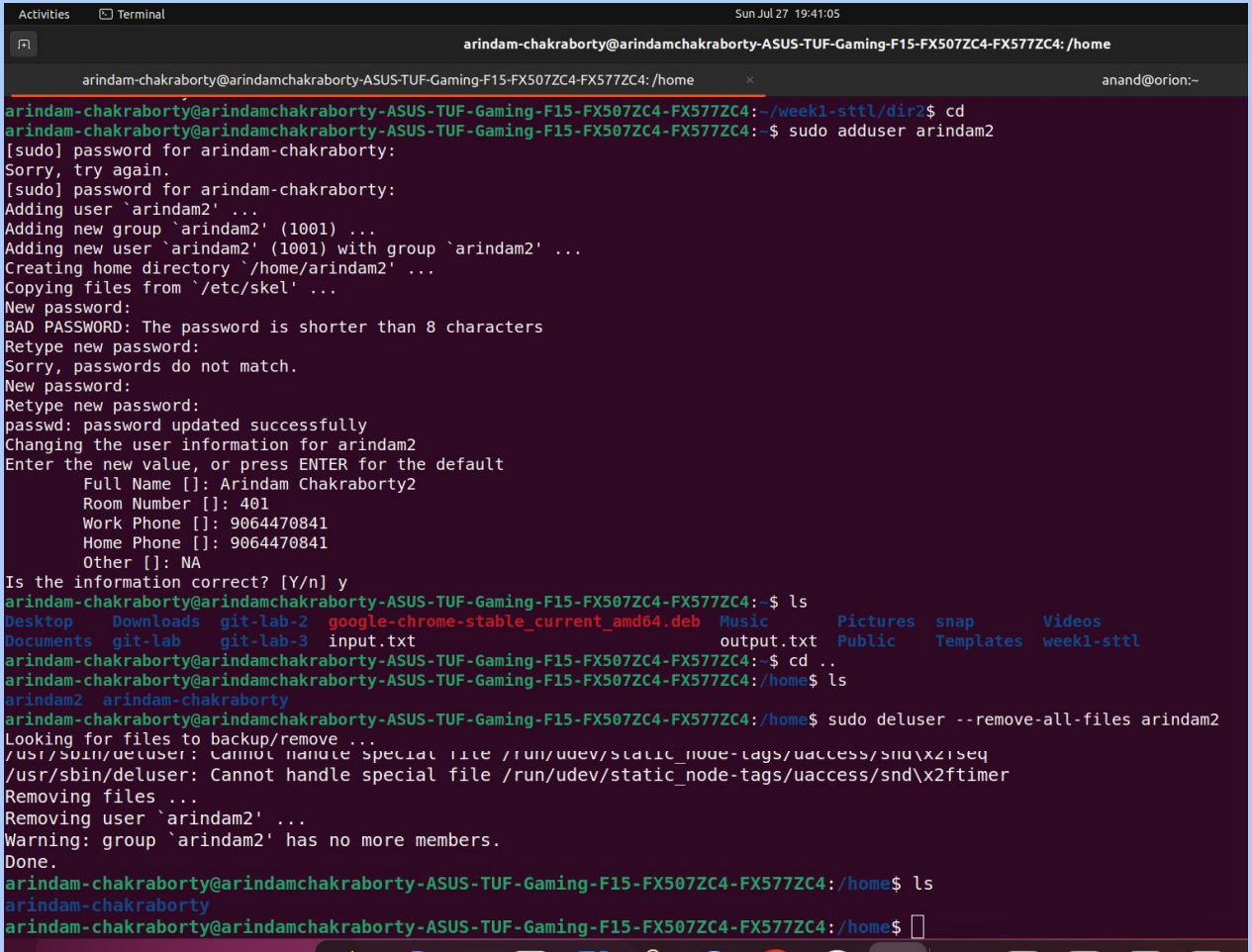


Deleting a permission from a file: sudo chmod [u/g/o] - [r/w/x] <file_name>
Adding a permission to a file: sudo chmod [u/g/o] + [r/w/x] <file_name>
Change ownership of file: sudo chown <user_name> <file_name>
Change group owner of file: sudo chgrp <group_name> file-5.txt

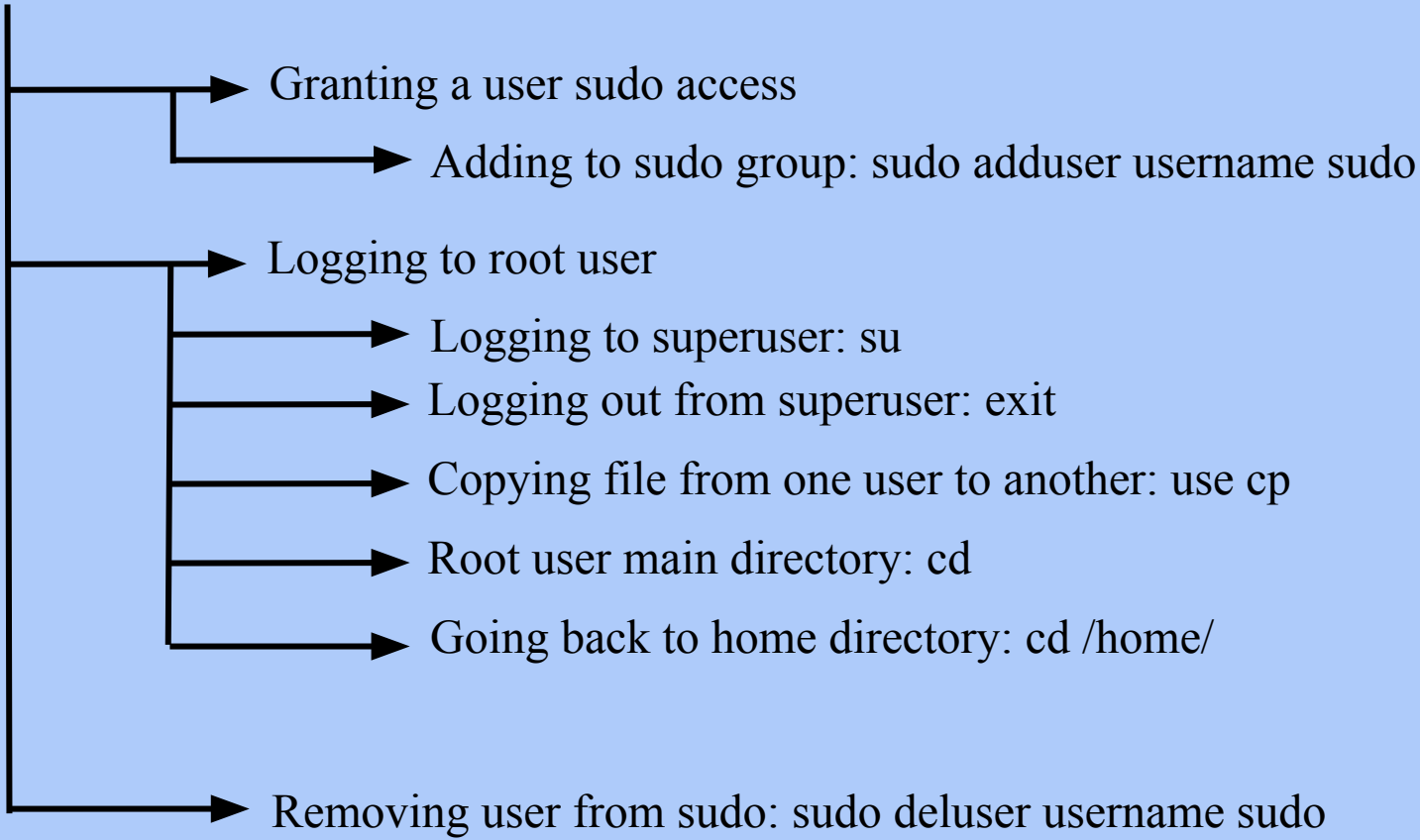
User and Administrative Management



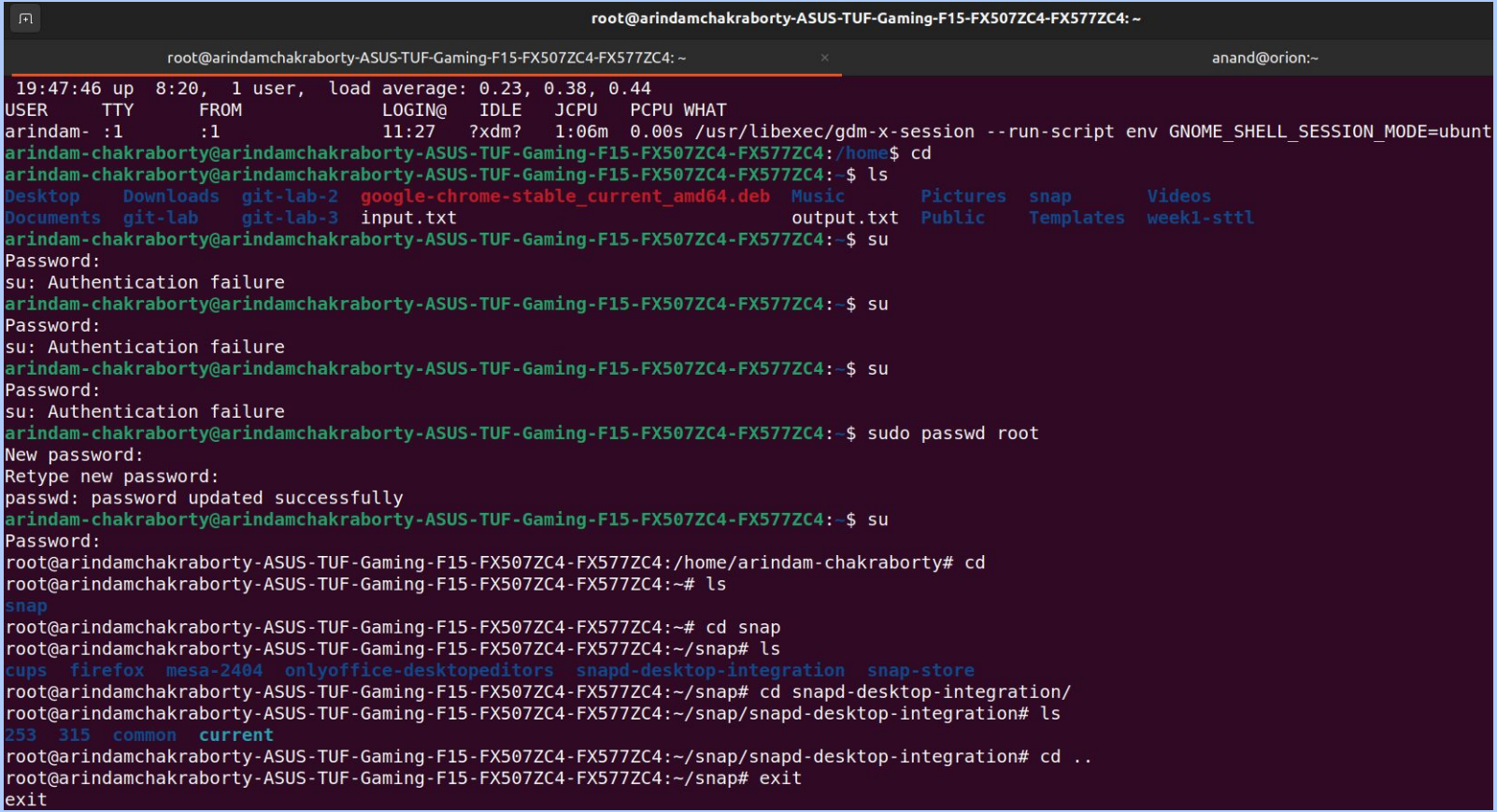
User and Administrative Management



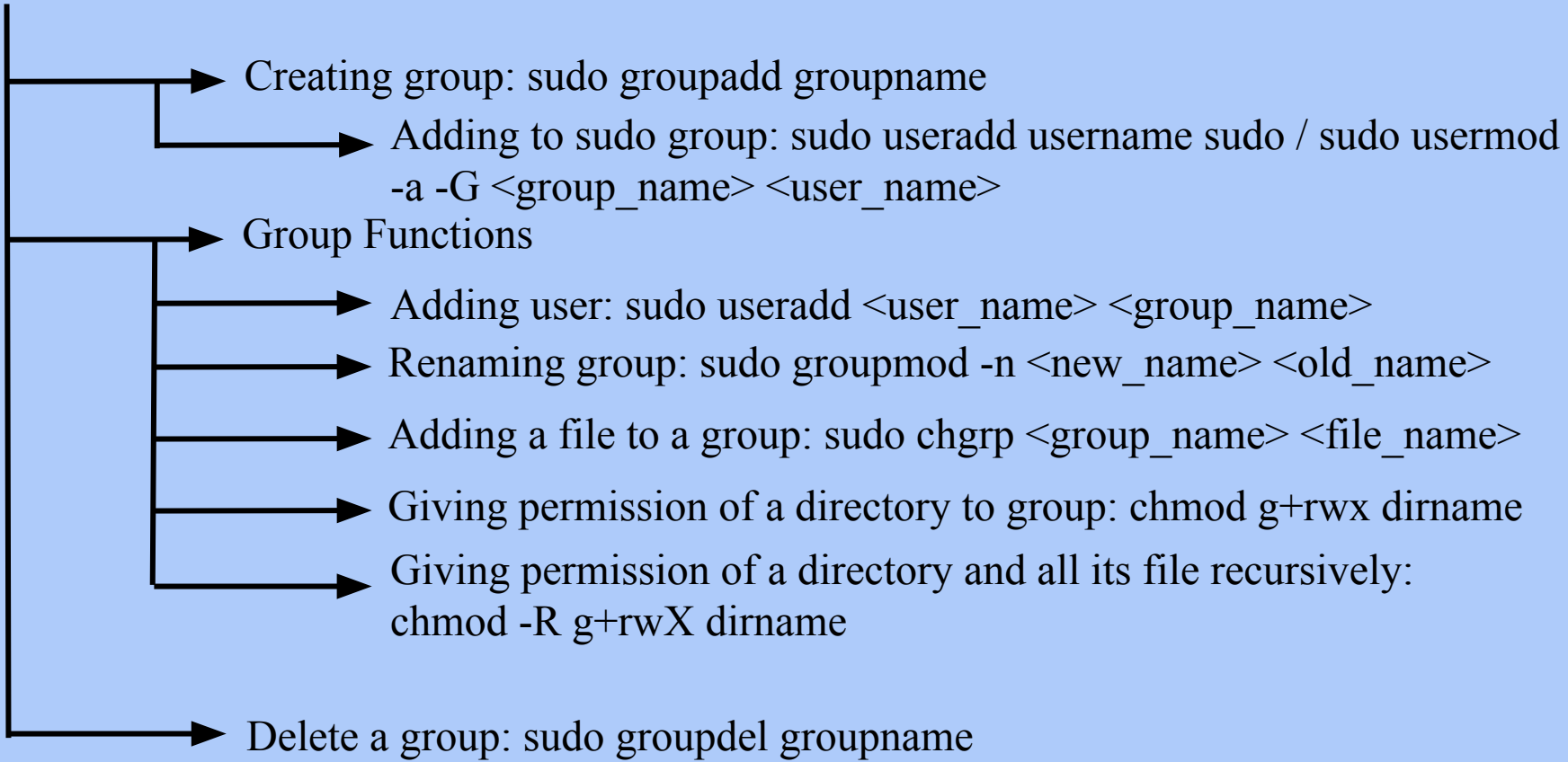
Administrative Privileges



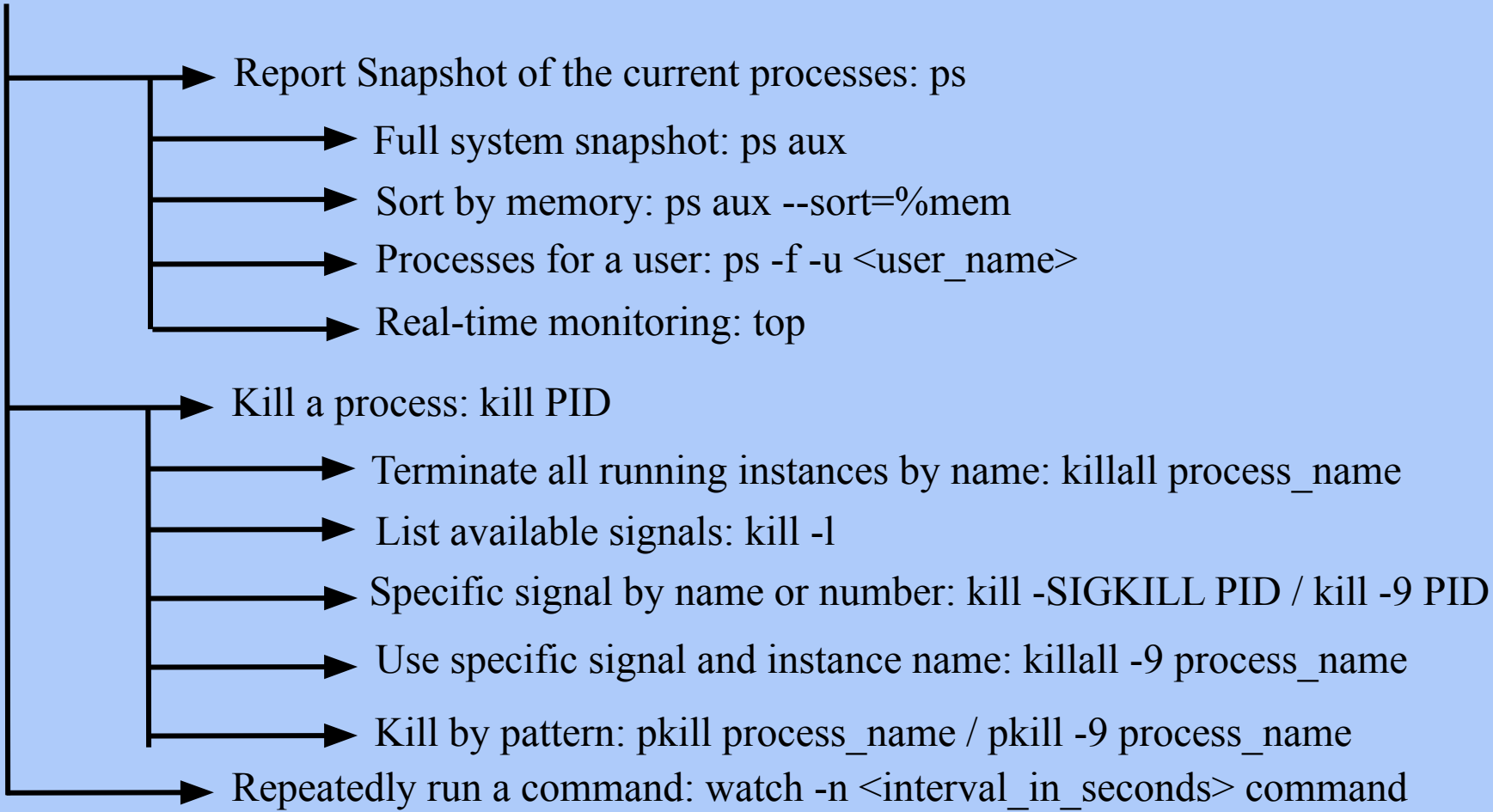
Admin Privileges



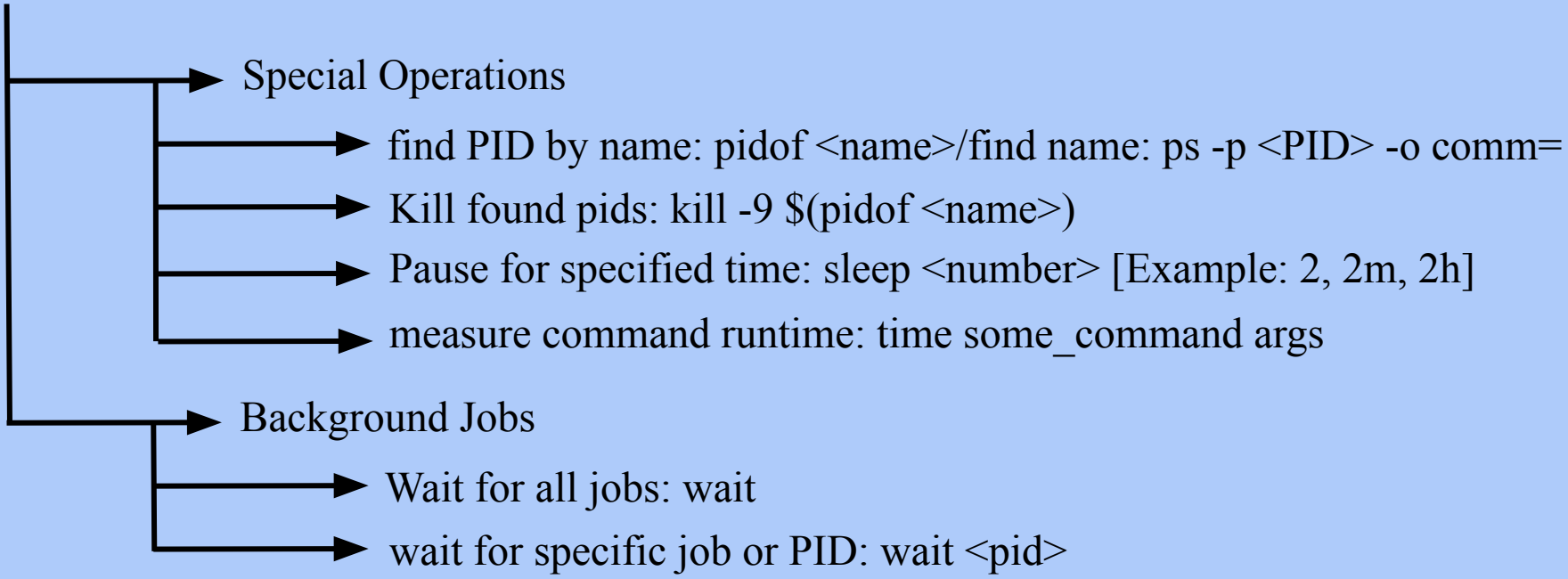
Group: (Extra)


Creating group: sudo groupadd groupname
Adding to sudo group: sudo useradd username sudo / sudo usermod -a -G <group_name> <user_name>
Group Functions
Adding user: sudo useradd <user_name> <group_name>
Renaming group: sudo groupmod -n <new_name> <old_name>
Adding a file to a group: sudo chgrp <group_name> <file_name>
Giving permission of a directory to group: chmod g+rwx dirname
Giving permission of a directory and all its file recursively: chmod -R g+rwX dirname
Delete a group: sudo groupdel groupname

Process Management Commands



Process Management Commands



Process Management Commands:

```
arindam+   17689  1.8  1.1 34338488 191000 ?     Sl   23:13   0:05 /snap/whatsapp-linux-app/2/whatsapp-linux-app --type=gpu-process --no-sandbo
arindam+   17694  0.3  0.4 34038492 74656 ?      Sl   23:13   0:01 /snap/whatsapp-linux-app/2/whatsapp-linux-app --type=utility --utility-sub-t
root       17695  0.0  0.0      0     0 ?        S    23:13   0:00 [nvidia-drm/timeline-42]
arindam+   17737 32.1  4.3 1191554288 688176 ?   Rl   23:13   1:35 /snap/whatsapp-linux-app/2/whatsapp-linux-app --type=renderer --enable-crash
root       17824  0.0  0.0      0     0 ?        I    23:14   0:00 [kworker/16:3-events]
root       17826  0.0  0.0      0     0 ?        I    23:14   0:00 [kworker/8:2]
root       17877  0.0  0.0      0     0 ?        I<   23:15   0:00 [kworker/u41:2-rb_allocator]
arindam+   17927  0.2  0.3 34018608 58720 ?      Sl   23:17   0:00 /snap/whatsapp-linux-app/2/whatsapp-linux-app --type=utility --utility-sub-t
root       17941  0.0  0.0      0     0 ?        I    23:17   0:00 [kworker/u40:3-ext4-rsv-conversion]
arindam+   17953  0.8  0.7 1461662928 112088 ?   Sl   23:18   0:00 /opt/google/chrome/chrome --type=renderer --crashpad-handler-pid=3283 --enab
root       17964  0.0  0.0  27308  4480 ?        S    23:18   0:00 /lib/systemd/systemd-udevd
arindam+   17965  0.0  0.0  12672  3360 pts/1    R+   23:18   0:00 ps aux
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~/week1-sttl$ ps -p 17694 -o comm=
whatsapp-linux-
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~/week1-sttl$ pidoff whatsapp-linux-
Command 'pidoff' not found, did you mean:
  command 'pidof' from deb sysvinit-utils (3.01-1ubuntu1)
Try: sudo apt install <deb name>
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~/week1-sttl$ pidof whatsapp-linux-
17927 17737 17694 17689 17647 17646 17572
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~/week1-sttl$ kill -9 $(pidof whatsapp-linux-)
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~/week1-sttl$ pidoff whatsapp-linux
Command 'pidoff' not found, did you mean:
  command 'pidof' from deb sysvinit-utils (3.01-1ubuntu1)
Try: sudo apt install <deb name>
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~/week1-sttl$ pidof whatsapp-linux
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~/week1-sttl$
```

Introduction to Text Processing Commands

- Print commands
  - Print string on the screen: echo “<string>”
  - Store in a file overwriting previous one: echo <string> > filename
  - Append line in the file: echo <string> >> filename
  - Storing output of some command to a file: command > file.txt (Overwriting)
  - Storing output of some command to a file: command >> file.txt (Without overwriting)
- Print file
  - Print the contents of a file: cat <file_name>
  - Print first n lines a file: head -n <num_lines> <file_name> [-c: first N bytes]
  - Print last n lines a file: tail -n <num_lines> <file_name>

Introduction to Text Processing Commands

```
arindam@NAMS-LAB-NTN-401-System-1:~$ touch file.txt
arindam@NAMS-LAB-NTN-401-System-1:~$ echo "This is STTL lab" > file.txt
arindam@NAMS-LAB-NTN-401-System-1:~$ cat file.txt
This is STTL lab
arindam@NAMS-LAB-NTN-401-System-1:~$ echo "Today is Thursday" > file.txt
arindam@NAMS-LAB-NTN-401-System-1:~$ cat file.txt
Today is Thursday
arindam@NAMS-LAB-NTN-401-System-1:~$ echo "This is STTL lab" > file.txt
arindam@NAMS-LAB-NTN-401-System-1:~$ echo "Today is Thursday" >> file.txt
arindam@NAMS-LAB-NTN-401-System-1:~$ cat fi
fib       fib.c     file.txt
arindam@NAMS-LAB-NTN-401-System-1:~$ cat file.txt
This is STTL lab
Today is Thursday
arindam@NAMS-LAB-NTN-401-System-1:~$ echo "Today is second lab session" >> file.txt
arindam@NAMS-LAB-NTN-401-System-1:~$ echo "Indian Institute of Technology, Bhilai" >> file.txt
arindam@NAMS-LAB-NTN-401-System-1:~$ echo "You are at Durg Station" >> file.txt
arindam@NAMS-LAB-NTN-401-System-1:~$ cat file.txt
This is STTL lab
Today is Thursday
Today is second lab session
Indian Institute of Technology, Bhilai
You are at Durg Station
arindam@NAMS-LAB-NTN-401-System-1:~$ head -n 2 file.txt
This is STTL lab
Today is Thursday
arindam@NAMS-LAB-NTN-401-System-1:~$ tail -n 3 file.txt
Today is second lab session
Indian Institute of Technology, Bhilai
You are at Durg Station
arindam@NAMS-LAB-NTN-401-System-1:~$ head -c 4 file.txt
Thisarindam@NAMS-LAB-NTN-401-System-1:~$ tail -c 7 file.txt
tation
arindam@NAMS-LAB-NTN-401-System-1:~$ tail -c 8 file.txt
Station
arindam@NAMS-LAB-NTN-401-System-1:~$
```


THANK
YOU